

The Impact of Hospital-Sponsored Health Plans on Readmission Rates

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PRELIMINARY

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Abstract

Hospital systems are increasingly vertically integrated (VI) with healthcare financing by sponsoring their own health plans. This paper studies whether admission to a (VI) hospital is associated with different 30-day unplanned readmission rate compared to standard non-VI hospitals. To address patient selection into hospitals, we develop an instrumental-variables design that leverages quasi-experimental variation in hospital assignment generated by (i) the ambulance transportation in emergency care episodes and (ii) the patient's relative distance to VI versus non-VI hospitals. We address unobserved hospital quality that may influence the decision to become a vertically integrated entity by including hospital fixed effects and year fixed effects, which allow us to exploit variation in hospitals' vertical integration status over time. We implement this identification strategy using administrative Medicare Advantage (MA) enrollment and claims data. We find that in less concentrated hospital markets the readmission probability at VI hospitals is about 3.4 percentage points lower than at other hospitals, and that the estimated treatment effect is smaller in markets with higher observed hospital concentration.

Keywords: Medicare Advantage, Vertical Integration, Provider-Sponsored Health Plan, Hospital Readmission, Hospital Competition.

1 Introduction

Provider-sponsored health plans (PSHPs)–Medicare Advantage (MA) plans sponsored by health systems that also deliver care–have become an increasingly important form of vertical integration in U.S. health care. In these arrangements, the same organization acts as both insurer and provider, potentially allowing it to internalize a larger share of the costs and benefits of clinical decisions. Proponents argue that tighter integration between financing and delivery may improve care coordination, strengthen discharge management, and reduce avoidable utilization. Critics emphasize that vertical integration may also increase market power, weaken competitive discipline, and alter incentives in ways that do not necessarily improve patient outcomes.

A natural setting in which to study these competing forces is 30-day unplanned readmission following emergency inpatient admission. Readmissions are costly for plans and for Medicare, are sensitive to discharge planning and care transitions, and are explicitly targeted by Medicare quality policy through the Hospital Readmissions Reduction Program (HRRP). In the context of hospital-sponsored plans, they are especially informative because integration may change incentives on both sides of the organization: hospitals may have stronger incentives to invest in discharge protocols and care coordination, while affiliated plans may have stronger incentives to manage patients after discharge.

The paper studies two distinct treatment margins. The first is *admission to a vertically integrated (VI) hospital*; the second is *enrollment in a VI plan*. Distinguishing these margins is important conceptually and empirically. If vertical integration primarily operates through hospital-side changes in clinical practice, discharge protocols, or provider-based care coordination, then admission to a VI hospital should matter even after accounting for plan enrollment. By contrast, if integration primarily operates through insurer-side care management, post-discharge monitoring, or network design, then enrollment in a VI plan should matter even holding the admitting hospital fixed. The empirical framework is designed to separate these channels as much as the data allow.

We also investigate whether the effect of vertical integration depends on the competitiveness of the local hospital market. There are strong reasons to expect such heterogeneity. A large literature in health economics and industrial organization shows that hospital performance and quality incentives depend on market structure. Competition can strengthen hospitals' incentives to invest in quality and operational performance, while high concentration can weaken those incentives or allow organizations with market power to capture rents without delivering commensurate quality improvements. ?, ?, and ? all emphasize the importance of competition for provider performance. Most directly, ? show that higher-priced

hospitals improve mortality only in less concentrated markets, whereas in concentrated markets higher prices raise spending without detectable quality gains. This literature provides a natural motivation for asking whether any quality advantage associated with vertical integration is larger in competitive hospital markets and weaker in highly concentrated ones.

Recent evidence on vertically integrated MA plans is suggestive but incomplete. Much of the literature focuses on plan characteristics, care experiences, or broad hospitalization patterns rather than readmissions from emergency episodes. ? show that integrated MA plans have higher star ratings and lower non-claims costs than non-integrated plans, although they also charge higher premiums. ? document the rapid growth of integrated MA plans and show that legacy-integrated plans tend to have higher star ratings and higher premiums than both non-integrated and newer integrated plans. On the patient-experience margin, ? find that integrated MA plans overall are not associated with meaningfully better care experiences than non-integrated plans, with the strongest advantages concentrated in legacy-integrated plans. Finally, ? study provider-integrated MA arrangements and find differences in inpatient care patterns and marginally lower mortality, but no difference in readmissions. Taken together, the existing evidence suggests that vertical integration may affect quality, costs, and care processes, but it leaves open whether hospital-sponsored plans reduce readmissions and, if so, whether that effect operates through the hospital side, the plan side, or both.

Turning from correlation to causal inference requires addressing two distinct endogeneity problems. First, patients may sort non-randomly across hospitals, so comparisons of readmissions between VI and non-VI hospitals may reflect unobserved patient severity, referral patterns, or hospital selection rather than the effect of integration itself. To address this concern, I exploit quasi-experimental variation in hospital assignment generated by emergency ambulance referral patterns and by patients' relative distance to VI versus non-VI hospitals. This design is implemented in emergency inpatient episodes, where patient discretion over the timing and location of care is limited. Second, beneficiaries may sort across MA plans, and VI-plan enrollment may be correlated with unobserved beneficiary or market characteristics. To address that margin, I use county-level variation in VI-plan availability together with a differential-distance shifter in a border-zip design. In the VI-plan regressions, I additionally include admitting-hospital fixed effects so that identification comes from comparing VI-plan and non-VI-plan enrollees treated at the same hospital.

The empirical results point to a clear asymmetry between hospital-side and plan-side vertical integration. In the preferred IV specifications, admission to a VI hospital is associated with a lower probability of 30-day unplanned readmission, and the estimated treatment effect is smaller in markets with higher observed hospital concentration. Evaluated at representa-

tive concentration levels, the implied effect is economically meaningful in less concentrated markets and becomes notably smaller in highly concentrated markets. By contrast, the VI-plan estimates are negative but imprecise and do not provide statistically precise evidence of an additional within-hospital plan-level spillover effect.

These findings have direct policy implications. First, they suggest that vertical integration may generate quality gains through the hospital-provider side of care, but that these gains are not uniform across markets. In relatively competitive hospital markets, integration may strengthen incentives to improve discharge management and reduce avoidable rehospitalizations. In more concentrated markets, however, the quality gains appear weaker. This implies that the welfare consequences of provider-sponsored plans cannot be evaluated independently of local competition policy. The same organizational form may generate meaningful coordination benefits in one market and much smaller benefits in another. From an antitrust and regulatory perspective, the results therefore support a more conditional view of vertical integration: integration may improve performance in settings where competitive discipline remains strong, but in highly concentrated markets its incremental quality benefits may be limited, strengthening the case for careful scrutiny of its market-power consequences.

This paper contributes to the literature on insurer-provider vertical integration in three ways. First, it studies a clinically meaningful and policy-relevant outcome—30-day readmissions—in a setting where care coordination is likely to matter. Second, it distinguishes between hospital-side and plan-side channels of vertical integration rather than treating integration as a single undifferentiated object. Third, it shows that the effect of vertical integration is heterogeneous across market structures, linking the analysis of integration to the broader literature on competition, quality, and hospital performance.

The remainder of the paper proceeds as follows. Section 2 describes Medicare Advantage, provider-sponsored plans, the emergency inpatient diagnoses in the sample, and the administrative data. Section 3 presents the endogeneity concerns and the instruments. Section 4 introduces the empirical specifications and reports the main results. Section 5 presents an illustrative model to rationalize the empirical findings. Section 6 concludes.

2 Institutional Setting and Data

This section describes the institutional environment of Medicare Advantage (MA), the role of provider-sponsored plans, and the data used in the empirical analysis. We first discuss the growth of vertically integrated provider-sponsored health plans and how these organizations combine payer and provider functions. We then explain why hospital readmissions provide a useful outcome for evaluating the effects of hospital-plan integration. Finally, we describe the administrative datasets used in the analysis and present descriptive statistics on readmission rates and patient episodes across different integration settings.

2.1 Medicare Advantage and Provider-Sponsored plans

Medicare Advantage (MA) is the alternative to traditional Traditional Medicare. Medicare advantage beneficiaries enroll in health plans offered by private insurance companies. Private plans are paid by Medicare a capitated amount per-beneficiary per month to provide coverage for TM Part A and Part B benefits. Plans are differentiated in terms of provider network and coverage, and negotiate with providers the payment rates for each service.

Provider-sponsored health plans (PSHPs) are becoming a prominent feature of the Medicare Advantage (MA) market. PSHPs are vertically integrated entities in which a provider organization owns and operates its own insurance plan to provide healthcare services. This integration combines the payer-side (insurance) and provider-side (hospitals, doctors) functions within a single organization. Common ownership may align incentives across payer and provider, reducing transaction costs and enabling clinical integration, potentially raising quality and efficiency. Yet integration can also enable patient steering and increased market power.

This vertically integrated organization often takes the form of a health system, but there are instances of health plans that are sponsored by provider organizations that do not include hospitals. In this paper we will exclusively focus on plans sponsored by health systems. We will refer to hospitals that sponsor a plan as vertically integrated hospitals and their sponsored plans as VI plans.

2.2 Hospital readmission and hospital-plan integration

This paper focuses on hospital readmissions as a clinically and financially meaningful measure to evaluate the impact of vertically integrated plans. Recent evidence indicates that a substantial share of acute admissions are followed by a rehospitalization within 30 days, with aggregate spending associated with readmissions on the order of tens of billions of

dollars annually in Medicare.¹ Readmissions are financially consequential for Medicare and hospitals, and they are explicitly targeted by Medicare’s Hospital Readmissions Reduction Program (HRRP) ².

In non-integrated arrangements, the insurer bears the marginal claims cost of rehospitalizations, while the hospital is largely reimbursed per admission and may not fully internalize downstream utilization costs that occur after discharge. When a hospital system sponsors an MA plan, the integrated entity captures a larger share of the surplus from preventing avoidable rehospitalizations, which strengthens incentives to invest in activities that require coordination across settings—for example, discharge planning, timely outpatient follow-up, medication reconciliation, and post-acute care management. Integration can also lower transaction and information frictions (e.g., shared data systems and unified care-management protocols), making these investments easier to implement at scale.

Readmissions are therefore an informative outcome for evaluating hospital–plan integration as they are widely viewed as a sentinel measure of care transitions and post-discharge management: they are sensitive to failures in communication, follow-up, and coordination with outpatient and post-acute providers, and they are financially consequential for both payers and hospitals. In addition, readmissions are a central target of Medicare quality policy through the Hospital Readmissions Reduction Program (HRRP), which explicitly frames communication and care coordination around discharge as channels for reducing rehospitalizations. These features make readmissions a natural metric for assessing whether PSHPs improve the efficiency of care coordination at the hospital–plan interface.

This focus is also consistent with the integrated-MA literature. Meyers, Mor, and Rahman (2020) motivate provider-integrated MA plans as a mechanism for improving coordination across payer and provider and evaluate differences in hospitalization processes and post-hospital outcomes including unplanned readmissions. Park, Langellier, and Meyers (2022) document that integrated MA plans differ from non-integrated plans along several dimensions, including the provision of supplemental benefits directly related to transitional care (e.g., post-discharge in-home medication reconciliation and readmission-prevention benefits). Related work on MA integration and beneficiary experiences emphasizes care coordination as a key dimension along which integrated arrangements may differ from non-integrated plans (Bejarano et al. 2024).

¹See, e.g., [Ziedan and Kaestner \(2021\)](#) for summary statistics and policy discussion.

²CMS evaluates hospital readmission performance over a multi-year window and applies payment reductions to hospitals with “excess” readmissions relative to national benchmarks. See [Gupta \(2021\)](#).

2.3 Data

Medicare Advantage Claims Data We use administrative data from the Centers for Medicare & Medicaid Services (CMS) for the years 2019 and 2022. The main data source for plan enrollment is the Master Beneficiary Summary File (MBSF), which contains individual-level data on a 5% random sample of Medicare beneficiaries. The MBSF records enrollment information for each beneficiary, including whether they are enrolled in Traditional Medicare or a Medicare Advantage (MA) plan.

The MBSF also includes detailed beneficiary characteristics such as age, gender, race, income measures, and county of residence. In 2019, total Medicare enrollment was approximately 61 million, so the 5% sample includes about 3 million individuals. Roughly 37% of beneficiaries were enrolled in MA plans. I exclude ESRD patients from the analysis and also special needs plans, pace plans, employer-sponsored plans or PFFS plans which have a different reimbursement scheme from CMS which is different from the standard plans.

The main data source for patients’ acute-care utilization is the Inpatient Encounter Data. These files identify the admitting hospital (CCN) and, for ambulance-transported episodes, the ambulance provider. They include ICD-10 diagnosis and procedure codes, length of stay, and discharge disposition, which we use to construct episode definitions, risk adjusters, and readmission outcomes. In addition, we use the Inpatient, Outpatient, Skilled Nursing Facility (SNF), Home Health Agency (HHA), Durable Medical Equipment (DME), and Carrier Encounter files to measure broader utilization and to construct spending-based measures in supplementary analyses.

Provider-sponsored health plans and VI hospitals. To identify integrated MA plans sponsored by health systems, we use the AHRQ Compendium of Health Systems, which contains the list of all health plans offered by hospitals and health systems. We link the hospitals in the Compendium to their CMS provider identification numbers (CCN) in order to merge the VI classification to hospital-level datasets (including the AHA Annual Survey for hospital characteristics). To identify health plans sponsored by other provider organizations, we additionally search through the full list of MA contracts and flag those whose sponsor is a provider organization. Throughout the descriptive analysis, we classify a hospital-year as vertically integrated, $VI_{ht} = 1$, if hospital h is affiliated with a provider-sponsored MA plan in year t .

Readmission measure We use the claims data to compute readmission at the individual level. A patient experiences an *unplanned* acute inpatient readmission if within 30 days of discharge from the *index* emergency admission, the patient is readmitted to an hospital for

the same condition. We treat multiple rehospitalizations within the 30-day window as a single event, consistent with HRRP-style targeted measures, which are most naturally interpreted as readmission probabilities rather than counts.³ Following CMS-style measurement conventions, we exclude same-day transfers and implement a planned-readmission algorithm to remove rehospitalizations that are predictably scheduled (e.g., selected elective procedures) rather than reflecting failures of post-discharge management.⁴ Because our data are Medicare Advantage encounters, we can track readmissions to *any* acute-care hospital, not only the index hospital.

2.4 Descriptive Statistics

We begin by presenting descriptive statistics on readmission rates and examining how they differ across vertically integrated plans and vertically integrated hospitals. Next, we report the share of episodes treated in vertically integrated settings, focusing on the distribution of patients across vertically integrated hospitals conditional on whether they are enrolled in a vertically integrated plan. Finally, we explore how these patterns vary across markets with different levels of concentration.

Readmission Rates Table 1 reports the mean and standard deviation of 30-day unplanned readmission rates by the vertical integration status of hospitals and plans. The overall average readmission rate in the sample is approximately 14.5%. Comparing hospitals, the average readmission rate is very similar between vertically integrated and non-vertically integrated hospitals (14.5% vs. 14.5%), suggesting little unconditional difference in readmission outcomes across hospital integration status. In contrast, there is a notice-

Table 1: Readmission Rates by Vertical Integration

VI Hospital	VI Plan		Total
	0	1	
0	0.147 (0.354)	0.121 (0.327)	0.145 (0.352)
1	0.149 (0.356)	0.132 (0.338)	0.145 (0.352)
Total	0.148 (0.355)	0.128 (0.334)	0.145 (0.352)

³Gupta (2021) notes that CMS does not differentiate between one and multiple readmissions within the 30-day period when constructing targeted readmission metrics.

⁴Ziedan and Kaestner (2021) describe the 30-day all-cause framework and common exclusions used in HRRP-style definitions.

able difference across plan integration status. Beneficiaries enrolled in vertically integrated plans have a lower average readmission rate (12.8%) compared to those enrolled in non-vertically integrated plans (14.8%). Looking at the interaction between hospital and plan integration status, the lowest readmission rate occurs when patients are enrolled in vertically integrated plans but treated in non-vertically integrated hospitals (12.1%), while the highest rate appears among patients in non-integrated plans treated at vertically integrated hospitals (14.9%). However, the differences across hospital integration categories remain small relative to the differences across plan integration status. Overall, these descriptive statistics suggest that plan-level vertical integration may be more strongly associated with lower readmission rates than hospital-level integration. But accounting for patient selection, case mix, or other confounding factors, we find that hospital-level vertical integration has more causal implications of patient readmissions.

Table 2: 30-Day Unplanned Readmission across Market Concentration

Market Concentration	Hospital VI		Plan VI	
	Non-VI	VI	Non-VI	VI
HHI < 1500	0.154 (0.361)	0.159 (0.366)	0.162 (0.368)	0.128 (0.334)
$1500 \leq \text{HHI} < 2500$	0.151 (0.358)	0.151 (0.358)	0.151 (0.358)	0.146 (0.353)
HHI ≥ 2500	0.141 (0.348)	0.137 (0.344)	0.142 (0.349)	0.123 (0.329)

Notes: Each cell reports the mean 30-day unplanned readmission rate with standard deviations in parentheses. Markets are categorized by county-level hospital HHI.

Table 2 reports the average 30-day unplanned readmission rates by hospital and plan vertical integration status across different levels of hospital market concentration. Several patterns emerge from the descriptive statistics. First, vertical integration at the plan level is consistently associated with lower readmission rates across all concentration categories. Second, differences in readmission rates by hospital vertical integration status are relatively small. As mentioned above, these descriptive patterns differ from the results obtained in our causal analysis, indicating the presence of patient selection and endogeneity.

Distribution of Patient Episodes Table 3 reports the distribution of episodes across the vertical integration status of plans and hospitals. The first number in each cell represents the frequency, while the number below reports the corresponding percentage. VI Hospital

1 represents any vertically integrated hospital, not necessarily a hospital integrated with the beneficiary’s plan, whereas VI Hospital 2 represents full integration, meaning that the hospital is vertically integrated with the beneficiary’s plan. Focusing on episodes associated

Table 3: Distribution of Episodes by Vertical Integration

VI Plan	VI Hospital			Total
	0	1	2	
0	127,381	58,978	0	186,359
	68.35	31.65	0.00	100.00
1	10,143	5,341	12,406	27,890
	36.37	19.15	44.48	100.00
Total	137,524	64,319	12,406	214,249
	64.19	30.02	5.79	100.00

with non-vertically integrated plans, about 68.4% of admissions occur in non-vertically integrated hospitals, while the remaining 31.6% occur in hospitals that are vertically integrated with some other plan. By construction, none of these episodes occur in hospitals integrated with the enrollee’s own plan.

Among episodes associated with vertically integrated plans, 36.4% occur in non-vertically integrated hospitals, 19.2% occur in hospitals integrated with another plan, and 44.5% occur in hospitals that are vertically integrated with the beneficiary’s own plan. This pattern suggests that vertically integrated plans tend to direct a substantial share of admissions to hospitals within their own integrated networks, although a significant portion of admissions still occurs outside the integrated hospital system.

Table 4: Distribution of Episodes across market concentration

Market Concentration	Hospital VI			Plan VI	
	0	1	2	0	1
$\text{HHI} < 1500$	57.36	33.98	8.66	84.28	15.72
$1500 \leq \text{HHI} < 2500$	58.24	33.34	8.42	85.56	14.44
$\text{HHI} \geq 2500$	67.76	27.98	4.26	88.19	11.81

Notes: Entries report the percentage distribution of episodes within each market concentration category. Hospital VI = 0 denotes non-integrated hospitals, Hospital VI = 1 denotes hospitals vertically integrated with some plan, and Hospital VI = 2 denotes hospitals vertically integrated with the beneficiary’s own plan. Plan VI = 0 denotes non-integrated plans and Plan VI = 1 denotes vertically integrated plans.

Table 4 reports the distribution of episodes by hospital and plan vertical integration status across different levels of hospital market concentration. The entries represent the percentage of episodes within each concentration category. Under the column group “Hospital VI”,

category 0 denotes non-vertically integrated hospitals, category 1 denotes hospitals vertically integrated with some plan (but not necessarily the beneficiary’s plan), and category 2 denotes hospitals vertically integrated with the beneficiary’s own plan. Under the column group “Plan VI”, category 0 represents non-vertically integrated plans and category 1 represents vertically integrated plans.

Several patterns emerge from the table. First, the share of episodes occurring in non-vertically integrated hospitals increases as market concentration rises. In unconcentrated markets ($\text{HHI} < 1500$), about 57.4% of episodes occur in non-integrated hospitals, whereas this share increases to 67.8% in highly concentrated markets ($\text{HHI} \geq 2500$). Correspondingly, the share of episodes in hospitals that are vertically integrated with some plan (category 1) declines as market concentration increases.

Second, episodes occurring in hospitals that are vertically integrated with the beneficiary’s own plan (category 2) are relatively rare across all market structures and become even less common in more concentrated markets. Their share falls from 8.7% in unconcentrated markets to 4.3% in highly concentrated markets.

Overall, these descriptive statistics suggest that both plan and hospital vertical integration are less prevalent in more concentrated hospital markets, with a larger share of episodes occurring in non-vertically integrated hospitals and plans as market concentration increases.

Diagnoses in the estimation sample The analysis focuses on emergency inpatient episodes for high-acuity, largely non-deferrable conditions, where patient discretion over the timing and location of care is limited and instrumental-variables strategies based on emergency transport and local plan availability are most credible. The sample spans a broad set of diagnosis categories. The most common groups are circulatory-system conditions (23.25%), infectious diseases (17.12%), injuries and trauma (13.57%), respiratory-system conditions (10.77%), and digestive-system conditions (8.86%). Additional episodes come from genitourinary conditions (6.87%), endocrine disorders (4.48%), and nervous-system conditions (4.00%), with the remaining 11.08% classified as other diagnoses. Thus, while the empirical design is built around emergency and non-deferrable admissions, the estimation sample is not limited to a single clinical condition; rather, it covers a broad range of acute diagnoses for which readmission is a meaningful indicator of inpatient quality and post-discharge care coordination.

3 Endogeneity and Instruments

This section discusses the two distinct endogeneity problems that arise in the empirical analysis and the instruments used to address them. The first concerns *admission to a vertically integrated hospital*, denoted H_i . Even in emergency settings, patients may be differentially routed to hospitals for reasons correlated with unobserved severity or local referral patterns. The second concerns *enrollment in a vertically integrated plan*, denoted P_i . Beneficiaries may sort into provider-sponsored plans based on unobserved health, preferences for integrated delivery systems, anticipated utilization, or differential access to providers. These unobservables can directly affect readmission risk and may therefore bias naive comparisons.

Accordingly, I use two excluded instrument sets: Z_i^H for the hospital-treatment margin and Z_i^P for the plan-enrollment margin. The same underlying geographic and institutional features generate plausibly exogenous variation in each case, but the identifying arguments differ depending on whether the endogenous variable is hospital assignment or plan enrollment.

3.1 Emergency hospital assignment and ambulance preferences

The key identifying idea for the hospital-treatment margin is that, in emergency episodes, the admitting hospital is heavily influenced by pre-hospital factors such as ambulance dispatch, ambulance availability, and patient location. Patients typically do not choose the ambulance company, and dispatch/rotation protocols together with mutual-aid agreements generate plausibly exogenous assignment to ambulance companies with heterogeneous hospital destination preferences (Doyle Jr et al., 2015). At the same time, ambulance companies often have stable tendencies to deliver patients to particular hospitals, reflecting historical routing patterns, contracts, operational convenience, or local familiarity.

Figure 1 illustrates the key source of quasi-random variation in hospital assignment. Two ambulances serve the same geographic area but differ in their propensity to transport patients to different hospitals. If one ambulance is systematically more likely to deliver patients to a VI hospital than another, then patients who are quasi-randomly assigned to those ambulances differ in their probability of admission to a VI hospital even though they face similar emergencies and originate from the same local area.

This motivates an ambulance-preference instrument for the hospital equation. Let $a(i)$ denote the ambulance company serving episode i , and let $d(i)$ denote the diagnosis group. I

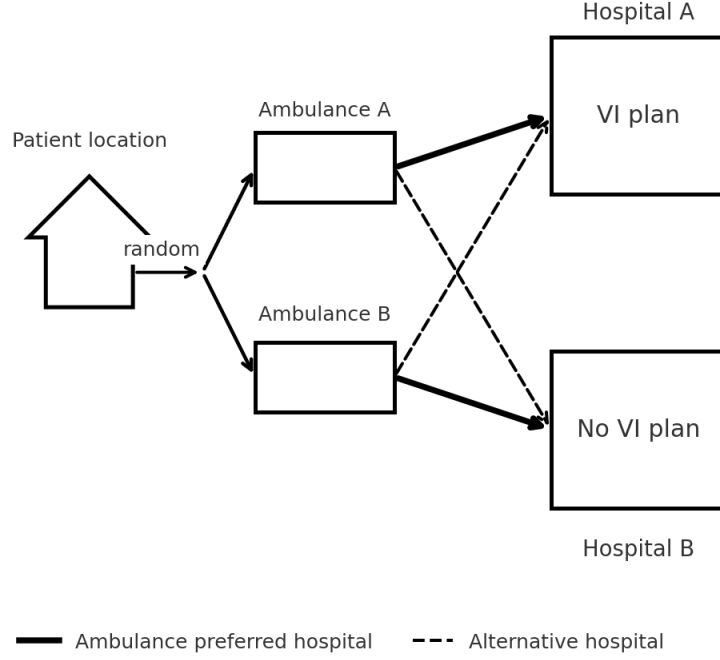


Figure 1: Hospital Assignment and Ambulance Preferences

construct the leave-one-out ambulance-level VI-hospital share:

$$Z_i^{A,H} = \frac{1}{N_{a(i),d(i),t(i)} - 1} \sum_{\substack{j \neq i: \\ a(j)=a(i), d(j)=d(i), t(j)=t(i)}} H_j.$$

Thus, $Z_i^{A,H}$ is the leave-one-out share of episodes with the same ambulance company, diagnosis group, and year that are admitted to a VI hospital. Conditioning on diagnosis is useful because ambulance routing patterns may differ across emergency conditions.

The identifying intuition is that, among otherwise similar patients facing similar emergencies, assignment to an ambulance with a higher propensity to transport patients to VI hospitals raises the probability of admission to a VI hospital while remaining plausibly orthogonal to patient unobservables conditional on location and the control set.

3.2 County-level plan availability and border variation

A naive comparison of readmissions between VI-plan enrollees and non-VI-plan enrollees is unlikely to be causal because beneficiaries may sort into provider-sponsored plans on unobserved dimensions. To address this concern, I first exploit the institutional fact that Medicare Advantage plans are offered at the county-year level: beneficiaries may enroll only

in plans filed for their county of residence. Thus, otherwise similar beneficiaries living in close geographic proximity but on different sides of a county border may face different menus of provider-sponsored plans.

Let $c(i)$ and $z(i)$ denote beneficiary i 's county and ZIP code, respectively. I compute a county-year PSHP availability index as the number of PSHP plans per thousand MA beneficiaries in county c and year t , scaled to lie in $[0, 1]$, and denote it by

$$Z_i^{A,P} \equiv \text{Avail}_{c(i),t(i)}.$$

The key identifying variation is that beneficiaries who live in the same ZIP code, or in ZIP codes very close to a county border, can face different county-level MA plan menus. I therefore restrict attention to ZIP codes that either span multiple counties or lie within a short distance of a county border.

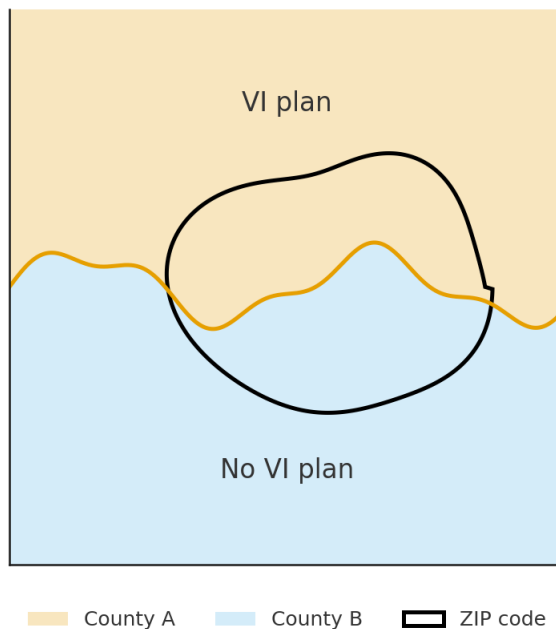


Figure 2: Within-ZIP variation in PSHP availability across county borders

Figure 2 illustrates this source of variation. Beneficiaries living in close geographic proximity may face different PSHP choice sets simply because they reside in different counties. This institutional feature generates plausibly exogenous variation in the feasibility of enrolling in a VI plan.

3.3 Differential distance as a common geographic shifter

I additionally use a geographic shifter based on relative proximity to VI and non-VI hospitals. Let $\text{dist}(r(i), h)$ denote distance from beneficiary i 's residence (or pickup location) $r(i)$ to hospital h . Define the nearest VI hospital $h^{VI}(i)$ and the nearest non-VI hospital $h^{NVI}(i)$, and construct

$$Z_i^{DD} \equiv \ln \text{dist}(r(i), h^{VI}(i)) - \ln \text{dist}(r(i), h^{NVI}(i)).$$

More negative values of Z_i^{DD} indicate greater relative proximity to a VI hospital.

This same differential-distance measure is useful for both endogenous margins, but with different interpretations.

First, in the hospital-assignment equation, Z_i^{DD} shifts the probability that an emergency episode is delivered to a VI hospital, because distance is a first-order determinant of hospital destination in emergency care.

Second, in the plan-enrollment equation, Z_i^{DD} shifts the relative attractiveness of enrolling in a provider-sponsored plan, because beneficiaries value proximity to in-network hospitals and providers. In that setting, proximity to a VI hospital system makes the affiliated plan more attractive.

Thus, the same geographic shifter enters both the hospital and plan instrument sets, but its identifying role differs depending on whether the endogenous regressor is hospital assignment or plan enrollment.

3.4 Excluded instrument

Combining the margins above, the excluded instrument set for admission to a VI hospital is

$$Z_i^H \equiv \left(Z_i^{A,H}, Z_i^{DD} \right).$$

The first component captures ambulance-specific destination preferences; the second captures relative geographic proximity to VI versus non-VI hospitals. The identifying assumption is that, conditional on the control set and fixed effects, the ambulance preference shifter and differential distance affect readmissions only through admission to a VI hospital. This assumption is most plausible in high-acuity emergency episodes, where patient discretion over the timing and location of care is limited.

The excluded instrument set for enrollment in a VI plan is

$$Z_i^P \equiv \left(Z_i^{A,P}, Z_i^{DD} \right).$$

The first component captures county-level PSHP availability; the second captures the relative convenience of proximity to a VI hospital system. For the plan-enrollment equation, the identifying assumption is that, conditional on beneficiary controls, plan characteristics, fixed effects, and admitting hospital, county-level PSHP availability and differential distance affect readmissions only through VI-plan enrollment. The hospital fixed effects in the plan regression are particularly important: they ensure that identification does not come from VI-plan enrollees being treated at systematically different hospitals on average. Rather, the design compares beneficiaries treated at the same hospital but facing different plan-enrollment incentives.

3.5 Estimation samples and sample construction

The estimation samples differ across the hospital and plan analyses because the identifying variation is different in each case. For the hospital-treatment margin, I restrict attention to high-acuity, effectively non-deferrable emergency conditions—such as acute myocardial infarction, stroke, and hip fracture—for which the index admission is largely outside the patient’s control. This sharpens the plausibility of the ambulance-based assignment design.

For the VI-plan analysis using the county-availability design, I restrict the sample to ZIP codes that either span multiple counties or lie within a short distance of a county border. This ensures that the identifying variation comes from geographically proximate beneficiaries facing different county-level MA plan menus. For specifications that use differential distance, I additionally require non-missing residential location information sufficient to compute distances. I report robustness to alternative border definitions and distance constructions in Appendix ??.

4 Empirical Framework and Results

The unit of observation is the beneficiary-by-episode, where an episode is defined as an inpatient admission originating in the emergency department (or an observation stay subsequently converted to inpatient) with no transfer-in from another acute-care facility. For each episode, I observe the beneficiary’s Medicare Advantage (MA) plan enrollment, the admitting hospital, and whether the beneficiary experiences an unplanned readmission within 30 days.

Let i index the beneficiary-episode. The primary outcome is an indicator for 30-day

all-cause unplanned readmission,

$$R_i = \begin{cases} 1 & \text{if beneficiary } i \text{ has an unplanned readmission within 30 days,} \\ 0 & \text{otherwise.} \end{cases}$$

Let $h(i)$ denote the admitting hospital, $p(i)$ the MA plan, and $t(i)$ the year. I study two treatment margins: (i) admission to a vertically integrated hospital and (ii) enrollment in a vertically integrated plan. Define

$$H_i \equiv \mathbb{1}\{\text{beneficiary } i \text{ is admitted to a VI hospital in year } t(i)\},$$

and

$$P_i \equiv \mathbb{1}\{\text{beneficiary } i \text{ is enrolled in a VI plan in year } t(i)\}.$$

To study heterogeneous effects, I interact both treatment indicators with the competitiveness of the local hospital market in which inpatient care is delivered. Let

$$M_{h(i)t} \equiv \ln(HHI_{c(h(i)),t}^{\text{hosp}}),$$

where $c(h(i))$ denotes the county of the admitting hospital. This is the relevant market characteristic because the mechanism of interest operates through the local provider market in which the hospital episode takes place. Importantly, I do not impose a common exogenous main effect of hospital-market concentration. Instead, I allow the relationship between concentration and readmissions to differ flexibly across treated and untreated groups. Accordingly, $M_{h(i)t}$ is best interpreted as an observed market characteristic along which the treatment effect may vary, rather than as a separately identified causal regressor.

4.1 Empirical Specifications

Effect of admission to a VI hospital. To estimate the effect of being treated at a VI hospital, I use a flexible interacted specification:

$$R_i = \beta_0 + \beta_H H_i + \gamma_0^H [(1 - H_i) M_{h(i)t}] + \gamma_1^H [H_i M_{h(i)t}] + X_i' \gamma + \alpha_{c(i)} + \tau_{t(i)} + \lambda_{d(i)} + \delta_{h(i)} + \phi_{p(i)} + \varepsilon_i, \quad (1)$$

where X_i is a vector of beneficiary and episode controls, $\alpha_{c(i)}$ are county fixed effects, $\tau_{t(i)}$ are year fixed effects, $\lambda_{d(i)}$ are diagnosis fixed effects, $\delta_{h(i)}$ are hospital fixed effects, and $\phi_{p(i)}$ are plan fixed effects in the specifications that include them.

This specification allows the slope on hospital-market concentration to differ flexibly

across non-VI and VI hospitals. Specifically, γ_0^H is the slope of $M_{h(i)t}$ for non-VI hospitals, while γ_1^H is the corresponding slope for VI hospitals. The economically relevant heterogeneity parameter is therefore the difference in slopes, $\gamma_1^H - \gamma_0^H$, which determines how the treatment effect of admission to a VI hospital varies with local hospital concentration.

Effect of enrollment in a VI plan. To estimate the effect of VI-plan enrollment, I use the analogous flexible specification:

$$R_i = \theta_0 + \theta_P P_i + \gamma_0^P [(1 - P_i)M_{h(i)t}] + \gamma_1^P [P_i M_{h(i)t}] + X_i' \eta + W_{p(i)t}' \rho + \alpha_{c(i)} + \tau_{t(i)} + \lambda_{d(i)} + \delta_{h(i)} + u_i, \quad (2)$$

where $W_{p(i)t}$ is a rich set of observed plan characteristics. Conditional on $\delta_{h(i)}$, the coefficient θ_P is identified from comparisons of VI-plan and non-VI-plan enrollees treated at the same hospital. This shuts down any mechanism that operates purely through hospital assignment (e.g. steering toward sponsor hospitals), so a negative θ_P would be consistent with a within-hospital plan effect such as plan-level care management or post-discharge coordination.

As in the hospital specification, this formulation allows the relationship between concentration and readmissions to differ flexibly across plan types. The slopes on $M_{h(i)t}$ for non-VI and VI plans are γ_0^P and γ_1^P , respectively, so the relevant heterogeneity parameter is the slope difference $\gamma_1^P - \gamma_0^P$.

I do *not* include plan fixed effects in (2). The reason is that VI status is effectively time-invariant within plan: a plan is either sponsored by a provider system or it is not. Including plan fixed effects would therefore absorb the main treatment indicator P_i and make the average VI-plan effect θ_P unidentified. Accordingly, identification of θ_P comes from instrument-induced variation in VI-plan enrollment across plans, conditional on admitting hospital, county, year, diagnosis, and observed plan characteristics.

A natural concern is that VI plans may differ from non-VI plans along unobserved quality dimensions that are not fully captured by $W_{p(i)t}$. Because plan fixed effects are infeasible, θ_P should not be interpreted as a within-plan effect of changing ownership status. Instead, the identifying assumption is that, conditional on controls and fixed effects, the excluded shifters Z_i^P are orthogonal to unobserved determinants of readmission. Under this interpretation, θ_P captures the effect of instrument-induced enrollment into the bundle of characteristics associated with a VI plan, holding the admitting hospital fixed.

4.2 Identification

Identification of the VI-hospital effect. Equation (1) contains three endogenous treatment-related objects: H_i , $(1 - H_i)M_{h(i)t}$, and $H_i M_{h(i)t}$. The first source of endogeneity is patient

selection into hospitals: unobservably healthier or sicker beneficiaries may be differentially admitted to VI hospitals. The second is hospital-level selection into VI status: hospitals that become vertically integrated may differ systematically from other hospitals. A third concern is that local hospital-market concentration may itself be correlated with unobserved determinants of readmissions. For this reason, I do not impose a common exogenous main effect of $M_{h(i)t}$. Instead, I allow the slope on concentration to differ flexibly across VI and non-VI hospitals.

To address these concerns, I instrument the treatment-related concentration components using the baseline hospital-assignment shifters and their interactions with $M_{h(i)t}$. Let

$$Z_i^H \equiv \left(Z_i^{A,H}, Z_i^{DD}, Z_i^{A,H} \times M_{h(i)t}, Z_i^{DD} \times M_{h(i)t} \right)$$

denote the excluded instrument set for the hospital specification. Intuitively, the same quasi-experimental variation that shifts the probability of admission to a VI hospital also shifts the concentration-weighted treatment components once interacted with the market moderator.

The fixed effects in (1) play complementary roles. Hospital fixed effects absorb all time-invariant hospital attributes, including baseline quality, discharge practices, coding intensity, and persistent post-acute referral patterns. County fixed effects absorb time-invariant local conditions that may affect both hospital choice and readmission risk. Year and diagnosis fixed effects absorb common shocks and differences in clinical composition. In the columns that include plan fixed effects, $\phi_{p(i)}$ also absorb time-invariant differences across plans. Thus, identification in the preferred VI-hospital specification comes from plausibly exogenous within-hospital variation over time in VI status, instrumented by Z_i^H .

Identification of the VI-plan effect. Equation (2) likewise contains three endogenous treatment-related objects: P_i , $(1 - P_i)M_{h(i)t}$, and $P_iM_{h(i)t}$. I instrument these using the county VI-plan availability index and the differential-distance shifter, together with their interactions with $M_{h(i)t}$. Let

$$Z_i^P \equiv \left(Z_i^{A,P}, Z_i^{DD}, Z_i^{A,P} \times M_{h(i)t}, Z_i^{DD} \times M_{h(i)t} \right)$$

denote the excluded instrument set for the VI-plan specification.

The identifying variation comes from plausibly exogenous shifts in the probability of VI-plan enrollment across plans serving the same beneficiary residential area, while conditioning on admitting hospital. Hospital fixed effects are crucial: they absorb time-invariant provider-side differences in quality, thereby ensuring that the plan effect is not identified from VI-plan enrollees being treated at better or worse hospitals on average. The flexible slope formulation

again avoids imposing a common exogenous main effect of local concentration. Instead, it allows the relationship between concentration and readmissions to differ across VI-plan and non-VI-plan enrollees.

A natural concern is that VI plans may differ from non-VI plans along unobserved quality dimensions that are not fully captured by $W_{p(i)t}$. Because plan fixed effects are infeasible, the parameter θ_P should therefore not be interpreted as a within-plan effect of changing ownership status. Instead, the identifying assumption is that, conditional on controls and fixed effects, the excluded shifters Z_i^P are orthogonal to unobserved determinants of readmission. Under this interpretation, θ_P captures the effect of being enrolled in the bundle of characteristics associated with a VI plan, holding the admitting hospital fixed. In other words, the design identifies the causal effect of instrument-induced enrollment into a VI plan, not a hypothetical effect of toggling the ownership status of a given plan while holding all other latent plan attributes fixed.

LATE interpretation. Under the standard IV assumptions of relevance, exclusion, and monotonicity, 2SLS identifies local average treatment effects (LATEs) for compliers whose treatment status is shifted by the excluded instruments. In the hospital specification, the excluded shifters Z_i^H identify the effect of admission to a VI hospital for beneficiaries whose hospital assignment is shifted by ambulance preferences and differential distance. In the plan specification, Z_i^P identify the effect of VI-plan enrollment for beneficiaries whose plan enrollment is shifted by county-level PSHP availability and differential distance.

Because the relationship between concentration and readmissions is allowed to differ flexibly across groups, the market-specific complier effects are

$$\tau_H(m) = \beta_H + (\gamma_1^H - \gamma_0^H)m \quad \text{and} \quad \tau_P(m) = \theta_P + (\gamma_1^P - \gamma_0^P)m.$$

Thus, the relevant heterogeneity parameters are the slope differences $\gamma_1^H - \gamma_0^H$ and $\gamma_1^P - \gamma_0^P$, which capture how the hospital and plan LATEs vary systematically with observed local hospital concentration. Importantly, these parameters should not be interpreted as the causal effect of changing concentration itself. Rather, they describe how the IV-identified treatment effects differ across markets with different observed concentration levels. Since $M_{h(i)t}$ is not mean-centered, the economically relevant objects are the marginal effects evaluated at representative HHI values.

4.3 Results

Table 5 reports the estimated effect of admission to a VI hospital on 30-day readmission rates. The preferred specifications are columns (1) and (2), which include hospital fixed effects and therefore exploit within-hospital variation over time in VI status. In both columns, the coefficient on H_i is negative, and the estimated slope of $\ln(HHI)$ is less negative for VI hospitals than for non-VI hospitals. Equivalently, the difference in concentration slopes, $\gamma_1^H - \gamma_0^H$, is positive, implying that the VI-hospital treatment effect is smaller in markets with higher observed hospital concentration.

Evaluating the market-specific treatment effect at representative HHI values yields economically meaningful magnitudes. At an HHI of 1000, admission to a VI hospital lowers the probability of 30-day readmission by roughly 3.5 percentage points. At an HHI of 2000, the implied reduction is about 2.5 percentage points, and at an HHI of 3000 the reduction is about 1.9 percentage points. Given average readmission rates around 14 percent, these are sizable effects. These calculations should be interpreted as the IV-identified treatment effect evaluated at representative observed levels of hospital concentration, rather than as the causal effect of changing concentration itself. The first-stage statistics are strong in all VI-hospital specifications.

Comparing columns (1) and (2), adding plan fixed effects leaves the point estimates very similar. Thus, the estimated VI-hospital effect is not driven by time-invariant differences across plans. By contrast, the coefficients become much smaller and statistically insignificant in columns (3) and (4), which exclude hospital fixed effects. This pattern indicates that controlling for time-invariant hospital heterogeneity is crucial. Overall, the evidence points to a robust hospital-side effect of VI on readmissions, with attenuation in more concentrated markets.

Table 6 reports the corresponding estimates for VI-plan enrollment. The preferred specifications are columns (1) and (2), which include hospital fixed effects and therefore compare VI-plan and non-VI-plan enrollees treated at the same hospital. In all four columns, the point estimate on P_i is negative, which is consistent with the hypothesis that VI-plan enrollment may reduce readmissions. However, the estimates are imprecise and statistically insignificant throughout. The interaction term $P_i \times \ln(HHI)$ is likewise not estimated precisely in any specification. As in the hospital specification, the relevant heterogeneity object is the difference in concentration slopes across VI-plan and non-VI-plan enrollees, $\gamma_1^P - \gamma_0^P$, which indicates whether the IV-identified plan effect varies across markets with different observed hospital concentration.

The appropriate interpretation of these null results is cautious. The VI-plan regressions do *not* provide statistically precise evidence of an additional within-hospital plan-level effect

Table 5: Effect of VI Hospitals

30-day Readm	(1)	(2)	(3)	(4)
VI Hospital	-0.1403** (0.0667)	-0.1403* (0.0801)	-0.0663 (0.0429)	-0.0663 (0.0516)
VI Hospital $\times$ ln HHI	0.0152** (0.0078)	0.0152* (0.0094)	0.0089 (0.0066)	0.0089 (0.0055)
Observations	171,354	172,176	171,388	171,388
Year FE	Yes	Yes	Yes	Yes
Diagnosis FE	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Plan FE	No	Yes	No	Yes
Hospital FE	Yes	Yes	No	No
SE cluster	Robust	Hospital	Robust	Hospital
First-stage F	71.473	1981.4	215.82	4046.5

Notes: Observations are weighted by the number of ambulance episodes and hospital admissions. Columns (1) and (2) include hospital fixed effects; columns (2) and (4) additionally include plan fixed effects. Standard errors are either heteroskedasticity-robust or clustered at the hospital level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6: Effect of VI Plan

30-day Readm	(1)	(2)	(3)	(4)
VI Plan	-0.4566 (0.4524)	-0.4566 (0.5312)	-0.4325 (0.3851)	-0.4325 (0.4723)
VI Plan $\times$ ln HHI	0.0288 (0.0375)	0.0288 (0.0482)	-0.0270 (0.0356)	-0.0270 (0.0463)
Observations	75,803	75,803	75,891	75,891
Year FE	Yes	Yes	Yes	Yes
Diagnosis FE	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Plan FE	No	No	No	No
Hospital FE	Yes	Yes	No	No
SE cluster	Robust	Hospital	Robust	Hospital
First-stage F	13.310	5.276	22.704	7.047

Notes: The sample is restricted to zip codes spanning two or three counties, and observations are weighted by the number of hospital admissions. Columns (1) and (2) include hospital fixed effects. Plan fixed effects are not included because VI status is time-invariant within plan and would absorb the main treatment indicator. Standard errors are either heteroskedasticity-robust or clustered at the hospital level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

of VI on readmissions. This is consistent with a mechanism in which the readmission gains associated with vertical integration operate primarily through the hospital/provider side

rather than through plan-level spillovers holding the admitting hospital fixed. At the same time, the imprecision of the estimates means that the data do not rule out modest plan-level effects. In particular, the clustered first-stage statistic in column (2) is weaker than in the VI-hospital specifications, and the border-zip design used to identify plan enrollment necessarily works with a more restricted sample.

Accordingly, I interpret Table 6 as showing that, conditional on admitting hospital and given the available identifying variation, there is no statistically precise evidence of a within-hospital VI-plan spillover effect. Combined with the stronger and more precise VI-hospital estimates in Table 5, the overall pattern is *more consistent* with a hospital-side channel than with a strong standalone plan-assignment channel, but it does not prove that the latter is exactly zero.

5 Illustrative model

This section presents a simple illustrative model to rationalize the three main empirical findings of Section 5. First, conditional on the treating hospital, readmission differences between patients enrolled in vertically integrated (VI) and non-VI plans are small. Second, comparing across hospitals, VI hospitals can exhibit lower readmission rates than otherwise similar standard hospitals. Third, this VI-hospital advantage weakens as hospital market power rises.

The key idea is that a VI hospital chooses a common level of effort to avoid readmissions that applies to all patients treated at the hospital, regardless of the insurance plan in which they are enrolled. This assumption captures the fact that many readmission-relevant investments—discharge planning, medication reconciliation, follow-up systems, and care-transition protocols—are hospital-wide organizational choices rather than patient-specific actions. As a result, patients treated at the same hospital face the same readmission technology, which naturally explains why within-hospital readmission differences by plan ownership may be limited.

At the same time, the incentive to choose effort depends on the composition of patients treated at the hospital and on the hospital’s payment environment. For patients enrolled in the hospital’s own VI plan, reducing readmissions saves the integrated system downstream medical costs under capitation. For patients enrolled in outside plans, incentives come instead from the hospital’s negotiated fee schedule. Vertical integration therefore changes effort through two channels: by internalizing the cost of readmissions for own-plan patients and by steering more own-plan patients into the VI hospital.

Patient types and market shares. Normalize the Medicare Advantage population to one. Let $\mu \in [0, 1]$ denote the share of beneficiaries enrolled in the VI plan. Let $s_0 \in [0, 1]$ denote the baseline probability that a beneficiary enrolled in a standard (non-VI) plan is treated at the VI hospital. We interpret s_0 as a reduced-form measure of the VI hospital’s market power or captive demand: higher s_0 means the hospital is more “must-have” even absent ownership of a plan.

Let s_1 denote the probability that a beneficiary enrolled in the VI plan is treated at the VI hospital. To capture steering, assume

$$s_1 = s_0 + h(1 - s_0)$$

where $h > 0$ capture the steering effect. Thus, stronger steering (higher h) increases the share of VI-plan patients treated at the VI hospital, while the steering effect $s_1 - s_0$ shrinks as s_0 increases. This feature is important: when the hospital is already very dominant, ownership of the plan has less room to change where patients are treated.

Two patient groups are treated at the VI hospital: (i) patients enrolled in the VI plan and treated at the VI hospital, with mass μs_1 , and (ii) patients enrolled in non-VI plans and treated at the VI hospital, with mass $(1 - \mu)s_0$. I abstract from VI-plan enrollees treated at other hospitals, since the motivation of the model is to explain differences in readmission outcomes between patients treated at VI and non-VI hospitals.

Readmission technology and effort cost. Let the hospital choose effort $e \geq 0$, interpreted as hospital-wide readmission containment effort. Readmission probability is

$$p(e) \equiv \Pr(R = 1 \mid e) = \bar{r} - e, \quad 0 \leq e \leq \bar{r},$$

cost of effort is

$$C(e) = \frac{c}{2}e^2, \quad c > 0.$$

We assume that VI hospitals and standard hospitals have the same readmission technology and the same cost of effort. Thus, the model aims to explain differences in readmission probability only through the financial incentives generated by plan ownership.

Payments, costs, and margins. For patients enrolled in non-VI plans, the hospital is paid negotiated fees. Let $F^{NR}(s_0)$ and $F^R(s_0)$ denote the hospital payment if the patient is *not* readmitted (NR) or *is* readmitted (R), respectively. These payments may depend on hospital market power, summarized by s_0 . The corresponding medical treatment costs are

w^{NR} and w^R , with

$$\Delta w \equiv w^R - w^{NR} > 0.$$

to capture that hospital costs are larger under readmission. Define hospital margins for non-VI-plan patients as

$$m^{NR}(s_0) \equiv F^{NR}(s_0) - w^{NR}, \quad m^R(s_0) \equiv F^R(s_0) - w^R.$$

The margin difference between a non-readmitted and a readmitted episode is

$$\Delta m(s_0) \equiv m^{NR}(s_0) - m^R(s_0) = \underbrace{(F^{NR}(s_0) - F^R(s_0))}_{\Delta F(s_0)} + \underbrace{(w^R - w^{NR})}_{\Delta w},$$

where

$$\Delta F(s_0) \equiv F^{NR}(s_0) - F^R(s_0).$$

A natural interpretation is that, when hospital market power is low, insurers can more easily penalize readmissions through the fee schedule, implying $\Delta F(s_0) > 0$. As the hospital becomes more powerful, insurers lose leverage, so the penalty embedded in negotiated fees weakens and $\Delta F(s_0)$ declines. Thus, throughout the discussion below, I allow

$$\Delta F(s_0) > 0, \quad \Delta F'(s_0) < 0,$$

Standard hospital. Consider a comparable standard hospital operating in a market with the same baseline share s_0 . This hospital treats only patients under negotiated-fee contracts, so its expected profit from one treated patient is

$$(1 - p(e))m^{NR}(s_0) + p(e)m^R(s_0).$$

If the hospital treats share s_0 of beneficiaries, it solves

$$\max_e \Pi^N(e; s_0) = s_0 \left[(1 - p(e))m^{NR}(s_0) + p(e)m^R(s_0) \right] - \frac{c}{2}e^2.$$

Using $p'(e) = -1$, the optimal non-VI effort is

$$e^N(s_0) = \frac{s_0 \Delta m(s_0)}{c} = \frac{s_0 (\Delta F(s_0) + \Delta w)}{c}. \quad (3)$$

VI hospital. Now consider the VI hospital. For patients enrolled in the VI plan and treated at the VI hospital, the integrated organization receives a capitated payment b from

Medicare and bears the medical costs of treatment. Since b does not vary with readmission status, the marginal benefit of effort for these patients comes entirely from lowering medical costs:

$$\frac{d}{de} \left[(1 - p(e))(b - w^{NR}) + p(e)(b - w^R) \right] = w^R - w^{NR} = \Delta w.$$

This is the mechanism through which vertical integration changes hospital incentives to avoid readmissions.

Thus, VI hospital expected profit on patients enrolled in its plan are

$$\pi_1(e) = b - \underbrace{s_1 [(1 - p(e))w^{NR} + p(e)w^R]}_{\text{beneficiaries treated at VI hospital}} - \underbrace{(1 - s_1) [(1 - p(e^O))F^{NR} + p(e^O)F^R]}_{\text{beneficiaries treated at other hospitals}}$$

where we assume that patients treated at other hospitals receive the level of effort e^O , and the VI-hospital profits depend on the fees negotiated between the VI hospital and the other hospitals.

For patients enrolled in non-VI plans and treated at the VI hospital, the marginal benefit of effort is the same as at a standard hospital, namely $\Delta m(s_0) = \Delta F(s_0) + \Delta w$. Thus, the expected profits on non-VI plans patient are given by

$$\pi_0(e) = s_0 [(1 - p(e))m^{NR}(s_0) + p(e)m^R(s_0)]$$

Therefore, adding of pieces together, the VI hospital solves

$$\max_e \Pi^{VI}(e; s_0) = \mu \pi_1(e) + (1 - \mu) \pi_0(e) - \frac{c}{2} e^2.$$

which gives the optimal VI-hospital effort

$$e^{VI}(s_0) = \frac{\mu s_1 \Delta w + (1 - \mu) s_0 (\Delta F(s_0) + \Delta w)}{c}. \quad (4)$$

Effort gap. Subtracting (3) from (4) yields

$$e^{VI}(s_0) - e^N(s_0) = \frac{\mu}{c} \left[(s_1 - s_0) \Delta w - s_0 \Delta F(s_0) \right] \quad (5)$$

Equation (5) shows that the effort gap between VI and standard hospitals has two components. The first term $(s_1 - s_0) \Delta w$ is the *internalization-and-steering gain*. Vertical integration makes readmission prevention more valuable because the hospital-system internalizes the downstream medical cost of readmissions for own-plan patients, and steering ensures that a larger fraction of those patients is treated at the VI hospital. This term is strongest

when steering is strong and when the VI hospital is not already dominant in the market.

The second term $s_0\Delta F(s_0)$ is the *fee-based incentive of the standard hospital*. When non-readmitted episodes are better reimbursed than readmitted episodes, standard hospitals already face a private incentive to reduce readmissions even without vertical integration. This compresses the VI advantage.

The VI hospital exerts more effort than the comparable standard hospital if and only if

$$h(1 - s_0)\Delta w > s_0\Delta F(s_0). \quad (6)$$

This condition has a clear economic interpretation: vertical integration increases effort when the extra gain from internalizing readmission costs on the additional own-plan patients steered into the VI hospital is larger than the fee-based incentive already faced by standard hospitals. Using (5), the readmission difference between the VI and standard hospital is

$$p^{VI} - p^N = -(e^{VI} - e^N)$$

Thus, whenever (6) holds, the VI hospital has lower readmission than the comparable standard hospital.

Effect of Market Concentration The size of the effort gap is affected by level of hospital market power. Define

$$H(s_0) \equiv h(1 - s_0)\Delta w - s_0\Delta F(s_0)$$

An interior threshold s_0^* , when it exists, solves $H(s_0^*) = 0$. For $s_0 < s_0^*$, the VI hospital exerts more effort and therefore has lower readmission rates than the standard hospital. For $s_0 > s_0^*$, the fee-based incentives of standard hospitals dominate and the effort gap becomes negative. If no interior threshold exists, then the VI advantage may remain positive over the relevant support of the data and simply shrink toward zero as s_0 rises.

As the hospital market share rises, the VI-hospital advantage in reducing readmissions becomes weaker in highly concentrated hospital markets.

$$s_1 - s_0 = h(1 - s_0) \rightarrow 0 \quad \text{as} \quad s_0 \rightarrow 1$$

In other words, hospitals that are already dominant in a market, owning the plan does little to bring additional patients in-house. Moreover, if insurers' bargaining power as s_0 increases, they are less able to impose strong fee penalties for readmissions, i.e.

$$\Delta F(s_0) \rightarrow 0 \quad \text{as} \quad s_0 \rightarrow 1,$$

Thus, both forces compress the difference in incentives between VI and standard hospitals, and the gap in readmission probability converges to zero.

Predictions. The model delivers two predictions that are consistent with our empirical findings:

- (i) If the *internalization-and-steering gain* is larger than the *fee-based incentive of the standard hospital*, hospitals that are plan sponsor have lower readmission than standard hospitals.
- (ii) Under the assumption that, as the VI-hospital market share increases, the incremental steering gain from plan-sponsoring shrinks, and $\Delta F(s_0)$ becomes smaller as insurer leverage weakens, the readmission gap diminishes as hospital market power increases.

Thus, the model suggests that vertical integration matters not because hospitals treat own-plan patients differently, but because ownership changes the hospital’s incentive to invest in common discharge and care-coordination protocols. These protocols benefit all patients treated at the hospital, which explains the limited within-hospital plan differences. At the same time, the benefit of integration is larger when ownership materially changes where patients are treated and when non-integrated hospitals do not already face strong payment penalties for readmissions.

6 Conclusion

This paper studies how hospital–plan vertical integration affects 30-day unplanned readmissions in Medicare Advantage. Provider-sponsored health plans combine payer and provider functions within a single organization, potentially allowing the integrated entity to internalize a larger share of the costs of avoidable readmissions and to improve care coordination across settings. At the same time, the consequences of vertical integration are unlikely to be uniform across markets: when hospital competition is weak, market power may blunt performance incentives and reduce the incremental value of integration.

Using Medicare Advantage enrollment and claims data, I study two distinct margins of vertical integration: admission to a vertically integrated hospital and enrollment in a vertically integrated plan. The empirical design is tailored to each margin. To study the hospital side, I exploit quasi-experimental variation in hospital assignment generated by ambulance referral patterns and differential distance to VI versus non-VI hospitals. To

study the plan side, I use county-level variation in VI-plan availability and differential-distance variation in a border-zip design, while conditioning on admitting hospital to isolate within-hospital plan comparisons.

The main results show a clear contrast between these two margins. Admission to a VI hospital is associated with a lower probability of 30-day readmission in the preferred IV specifications, and the estimated treatment effect is smaller in markets with higher observed hospital concentration. These effects are economically meaningful at moderate levels of concentration and become smaller in highly concentrated markets. By contrast, the VI-plan specifications yield negative but statistically imprecise estimates and do not provide precise evidence of an additional within-hospital plan-level spillover effect.

Taken together, the results suggest that the readmission benefits of vertical integration in this setting operate primarily through the hospital/provider side rather than through a strong standalone plan-assignment channel. In other words, vertical integration appears to matter more through where and how patients are treated than through enrollment in an integrated plan holding the hospital fixed. At the same time, the imprecision of the VI-plan estimates means that modest plan-level spillovers cannot be ruled out.

The analysis also points to an important interaction between integration and competition. The evidence is more consistent with vertical integration improving hospital performance in less concentrated markets than in highly concentrated ones. This pattern echoes a broader literature showing that hospital quality and performance can depend on the competitive environment and suggests that the consequences of integration cannot be understood separately from local market structure. (???)

To address concerns that hospitals may endogenously adopt VI status in response to pre-existing quality trends, I supplement the main MA analysis with a hospital-level FFS validation exercise. An event study around first observed VI adoption finds no evidence of differential pre-trends in the included pre-adoption lead years. While this exercise is not itself the main source of identification, it reduces concern that the hospital-side estimates merely reflect pre-existing improvements in performance among eventual adopters.

These findings have two broader implications. First, they suggest that vertical integration may generate efficiency gains through improved inpatient and discharge-related care processes, but that these gains are not guaranteed to be equally strong across market structures. Second, they caution against treating “vertical integration” as a single mechanism: hospital-side and plan-side channels can differ materially in both magnitude and precision. From a policy perspective, the results therefore point toward a more conditional view of integration. The welfare consequences of hospital-sponsored plans are likely to depend not only on whether hospitals and plans are jointly owned, but also on the competitive environment

in which that integration takes place.

Several questions remain for future work. One is whether the hospital-side effects documented here extend to other outcomes, such as post-acute utilization, spending, or mortality. Another is whether plan-side effects become more apparent in settings with richer plan-level variation or broader samples than the border-zip design used here. More generally, understanding how vertical integration interacts with local competition remains central for evaluating both the efficiency and the market-power consequences of the continued expansion of provider-sponsored plans in Medicare Advantage.

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A Appendix A

A.1 Validation

A remaining concern for the interpretation of hospital-level vertical integration (VI) effects is that the decision to become a plan sponsor may itself be endogenous. In particular, hospitals may adopt VI status as part of a broader organizational change—for example, improvements in care management, discharge planning, or system integration—that begins before formal adoption. If so, lower readmission rates at hospitals observed as vertically integrated could partly reflect pre-existing trends rather than the causal effect of VI status itself. This concern is distinct from patient-level selection into hospitals: even if our instruments plausibly address patient assignment, and hospital fixed effect net out time invariant differences across hospitals, the hospital’s own adoption of a sponsored plan may still be correlated with unobserved changes in outcomes.

To assess this threat, I use a separate hospital-level validation exercise based on fee-for-service (FFS) hospital-episode-year data from 2016–2023. These data are not used as the main source of identification for the Medicare Advantage analysis. Instead, they provide a useful validation setting because they allow direct observation of hospitals’ VI status over time and yearly readmission outcomes for a distinct patient population. The maintained idea is that if hospitals that later adopt VI were already on differential hospital-wide readmission trajectories before adoption, such trends should plausibly also be visible in the FFS data.

Because the timing of adoption varies across hospitals, I center the analysis around each hospital’s *first* observed year of VI adoption. Let G_h denote the first year hospital h becomes vertically integrated, and define event time as $k = t - G_h$. I then estimate the following event-study specification:

$$\ln(\text{RR}_{het}) = \beta \ln(\text{ERR}_{het}) + \sum_{k \neq -1} \theta_k \mathbb{1}\{t - G_h = k\} + X'_{ht}\gamma + Z'_{mt}\delta + \alpha_{he} + \lambda_{et} + u_{het}, \quad (7)$$

where RR_{het} is the observed FFS readmission rate for hospital h , episode e , and year t ; ERR_{het} is the expected readmission rate for that hospital-episode-year cell; X_{ht} denotes time-varying hospital controls; Z_{mt} denotes time-varying market controls; α_{he} are hospital-episode fixed effects; and λ_{et} are episode-year fixed effects. Standard errors are clustered at the hospital level. The omitted event-time category is $k = -1$, so each θ_k measures the difference in log observed readmission rates at event time k relative to the year immediately before first adoption.

This specification is intentionally demanding. Hospital-episode fixed effects absorb all time-invariant differences across hospital-episode cells, while episode-year fixed effects absorb

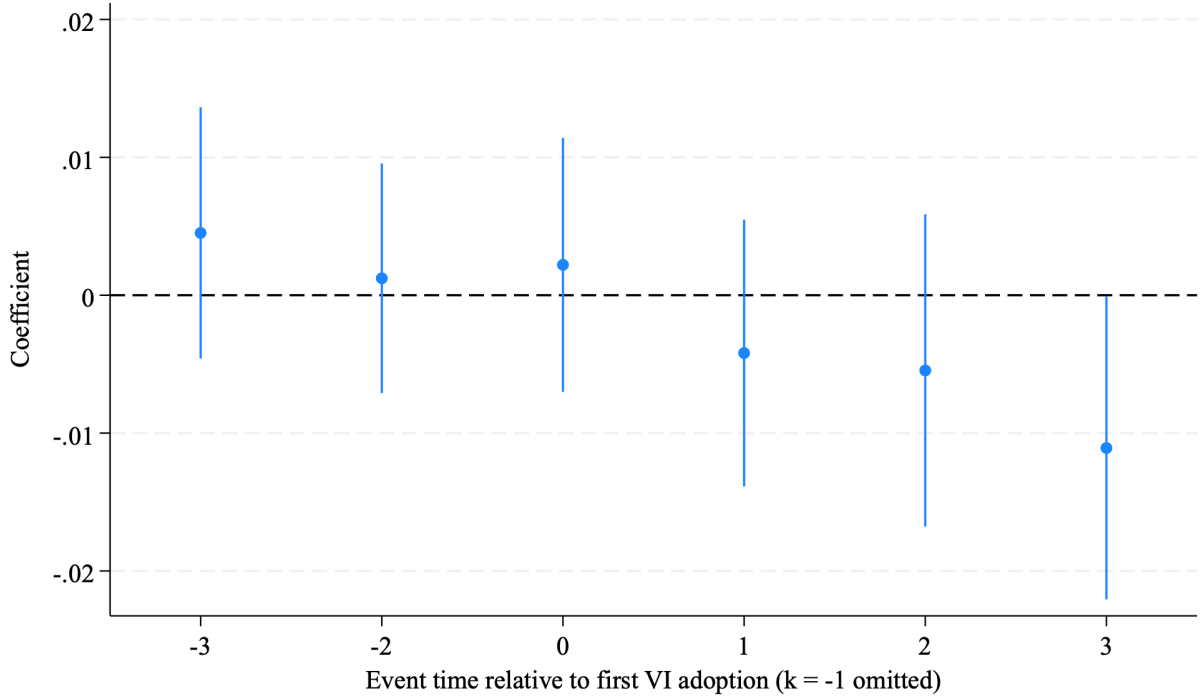


Figure 3: FFS event study around first hospital VI adoption

nationwide shocks that vary by episode over time. Conditional on expected risk, hospital controls, market controls, and these high-dimensional fixed effects, the lead coefficients θ_k for $k < 0$ provide a direct test of whether hospitals that later adopt VI were already on different outcome trajectories before first adoption. In this sense, the event study serves as a validation exercise for the hospital-adoption margin rather than as the primary source of causal identification. The coefficients of main interest are therefore the pre-adoption leads, while the post-adoption lags are interpreted more cautiously as descriptive dynamic evidence.

Figure 3 plots the estimated event-time coefficients. The pre-adoption coefficients are small in magnitude and statistically imprecise. In the baseline event-study specification, the coefficients at $k = -3$ and $k = -2$ are approximately 0.0045 and 0.0012, respectively, with neither individually distinguishable from zero. More importantly, a formal joint test of the included lead coefficients fails to reject the null that they are jointly zero ($F = 0.55$, $p = 0.579$). Thus, within the observed pre-adoption window, the FFS data provide no evidence that future adopters were already on differential readmission trends before first VI adoption.

The post-adoption coefficients are negative and become more pronounced over time. The contemporaneous coefficient at $k = 0$ is close to zero, while the estimates at $k = 1$, $k = 2$, and $k = 3$ are negative, with the largest decline appearing three years after first adoption. In

the current specification, the estimate at $k = 3$ is approximately -0.011 and is statistically significant at conventional levels. Because the purpose of this exercise is primarily to evaluate pre-trends, I do not emphasize these lag coefficients as causal treatment effects. Nevertheless, their sign is consistent with lower readmission rates after adoption.

Taken together, the event-study validation yields two conclusions. First, the absence of statistically detectable pre-adoption leads weakens the concern that hospitals that later adopt VI were already experiencing systematically different readmission trends before adoption. Second, the post-adoption coefficients move in the expected direction, consistent with improved outcomes after adoption, although those dynamics are interpreted as descriptive rather than definitive causal evidence. Overall, this FFS validation exercise supports the view that the main Medicare Advantage findings are unlikely to be driven solely by pre-existing hospital-level trends in readmission performance.